Introduction of Programming

Lecture 28

Virtual University

Today's Lecture

- How memory allocation is done in C++
- How is it different from C style
- Advantages of memory allocation in C++
- Uses of memory allocation
  - Classes
  - Objects

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Pointers to a data structure

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Pointers to a class

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->

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Memory Allocation

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malloc ( ) ;

calloc ( ) ;

realloc ( ) ;

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```
malloc ( 10 * ( sizeof ( int ) ) ) ;
```

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free ( )

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new

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new int ;

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Example

```
int * iptr ;
iptr = new int ;
```

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Example

```
new char ;
new double ;
```

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delete

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Example

```
int * iptr ;
iptr = new int ;
delete iptr ;
```

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Example

```
int * iptr ;
iptr = new int [ 10 ] ;
```

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Example

```
new data_type [ Number_of_locations ] ;

new double [ 10 ] ;
```

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Example

```
int *iptr ;
iptr = new int [ 10 ] ;
delete iptr ;
```

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Example

Date *dptr ;

dptr is a pointer to an object of type date

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```
dptr = new Date ;
```

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Example

```
main ( )
{
    Date mydate ;
    cout<< sizeof ( mydate ) ;
}
```

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Example

```
int *iptr ;
iptr = new int [ 10 ] ;
```

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Example

```
Date *dptr ;
dptr = new Date [ 10 ] ;
```

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Example

```
Date date1 , *dptr ;
date1.setDate ( ) ;

dptr = new Date ;
dptr ->setDate ( );

dptr.setDate ( ) ;
```

Wrong

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Protected

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Destructor

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- Allocate enough space for the new data
- Populate that space
- Delete the previous space
- Point the new space to the pointer pointing to the original data

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```
char *name =new char [ string_length ]
```

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delete [ ] name

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delete [ ] pointer_name

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Example

```
main ( )
{
  Date mydate ( “01-12-2002” ) ;
  mydate.display ( ) ;
}
```

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Messages

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Method

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